

Insertion Sort

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Key
44

[0]	[1]	[2]	[3]	[4]	[5]
55	44	22	66	11	33

j i

44 < 55 ?? Yes
1) Shift 55 one step ahead
2) Decrement j

[0]	[1]	[2]	[3]	[4]	[5]
	55	22	66	11	33

j i

As we decrement j, we realised that j has reached -1. hence there is no element, left to compare with 44.
So, copy 44 to index j+1

[0]	[1]	[2]	[3]	[4]	[5]
44	55	22	66	11	33

Iteration 2:

Key
22

[0]	[1]	[2]	[3]	[4]	[5]
44	55	22	66	11	33

j i

Is j at valid index ? Yes
AND
22 < 55 ? Yes
1) Shift 55 1 place ahead
2) Decrement j

[0]	[1]	[2]	[3]	[4]	[5]
44		55	66	11	33

j i

Is j at valid index ? Yes
and
22 < 44 ? Yes
1) Shift 44 one place ahead
2) Decrement j

[0]	[1]	[2]	[3]	[4]	[5]
	44	55	66	11	33

j i

As j has reached -1 index,
There is no element left to
Compare with 22, hence copy
22 to j+1 index

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

Key
66

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

j i

Is j at valid index ? Yes
And
66 < 55 ? No
Copy key to j+1 index

Key
11

[0]	[1]	[2]	[3]	[4]	[5]

Is j at valid index ? Yes
And

Key
11

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

j i

Is j at valid index ? Yes
And
 $11 < 66$? Yes
1) Shift 66 one step ahead
2) Decrement j

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55		66	33

j i

Is j at valid index ? Yes
And
 $11 < 55$? Yes
1) Shift 55 one step ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
22	44		55	66	33

j i

Is j at valid index ? Yes
And $11 < 44$? Yes
1) Shift 44 one step ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
22		44	55	66	33

j i

Is j at valid index ? Yes
And
 $11 < 22$? Yes
1) Shift 22 one step ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
	22	44	55	66	33

j i

Is j at valid index ? No
Copy 11 to j+1 index

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55	66	33

Key
33

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55	66	33

j i

Is j at valid index ?
 $J \geq 0$? Yes
And
 $33 < 66$?
Key < arr[j] ? Yes
1) Shift 66 one place ahead
Shift arr[j] to arr[j+1]
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55		66

j i

$J \geq 0$? Yes
Key < arr[j] ? Yes
1) Shift 55 to j+1 index
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44		55	66

j

i

J >= 0 ? Yes
key < arr[j] ? Yes
1) Shift arr[j] to j+1 index
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
11	22		44	55	66

j

i

j >= 0 ? Yes
And
Key < arr[j] ?
33 < 22 ? No

[0]	[1]	[2]	[3]	[4]	[5]
11	22	33	44	55	66

Copy key to j+1 index.

Check 2 conditions
If both the conditions are true, perform 2 actions
If any one condition is false, copy the key to j+1 index